

Data Processing Report

Construction of a Balanced Multi-Label Dataset for Pulmonary Adverse Drug Reactions

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1 Introduction

1.1 Objective

The primary goal of this pipeline is to assemble a labeled dataset to support machine learning modeling for the prediction of four specific pulmonary adverse drug reactions (ADRs):

1. **Interstitial Lung Disease (ILD) / Fibrosis**
2. **Pulmonary Edema**
3. **Pleural Effusion / Disorders**
4. **Pulmonary Embolism**

The final dataset is designed to be balanced, containing both positive instances (drugs known to cause these ADRs) and negative instances (drugs with no evidence of these ADRs in extensive biomedical databases).

1.2 Pipeline Overview

The pipeline integrates data from four distinct curated databases.

2 Data Sources Description

The pipeline integrates data from four distinct databases. These raw datasets were collected from public sources and subsequently processed through MEHC Curation to ensure data quality and consistency.

Source File	Origin	Type of Data	Specific Role in Pipeline
adrecs_pneumotox_curated.csv	ADReCS & PNEUMOTOX	Hierarchical patterns (e.g., I.a, II.d) and textual descriptions.	Provides hierarchical classification. The Pattern column contains codes that map to pulmonary toxicity phenotypes.
sider_respiratory_adrs.csv	SIDER (Side Effect Resource)	Marketed drugs with pre-extracted binary labels for the 4 specific ADRs.	Provides a large volume of real-world marketed drugs filtered specifically for the 4 respiratory side effects, directly contributing to the positive sample pool.
drugcentral_smiles.csv	DrugCentral (Open-access drug database)	INN and canonical SMILES strings.	Structural augmentation. Adds molecules to increase chemical diversity, primarily serving as candidates for the negative set.
ctd_mechanism_markers.csv	CTDbase (Comparative Toxicogenomics Database)	Chemical-gene-disease interaction data.	Identifies compounds annotated as mechanistic markers related to the 4 target ADR patterns, enriching the positive sample pool with biologically validated molecules.

Table 1: Summary of data sources used in the pipeline

3 Detailed Process

3.1 Step 1: Integration of Primary Labeled Sources

Script	Action	Output
01_merge_adrecs_sider.py	Concatenates the ADReCS/PNEUMOTOX dataset with the SIDER dataset.	01_merged_adrecs_sider.csv

Table 2: Step 1: Merging primary labeled sources

3.2 Step 2: Label Standardization via Pattern Parsing

Script	Action	Output
02_annotate_labels_from_pattern.py	Extracts canonical codes (e.g., I.b, V.a, VI.a) from the <code>Pattern</code> column using regex. Also extracts <code>ADReCS</code> : text clauses.	02_labeled_by_pattern.csv

Table 3: Step 2: Label standardization

Label Mapping Rules:

- **ILD:** Assigned if code is I.b, I.g, or text contains `interstitial/fibrosis`.
- **Edema:** Assigned if code is II.a, II.d, or text contains `pulmonary edema`.
- **Pleural:** Assigned if code is V.a, or text contains `pleural effusion`.
- **Embolism:** Assigned if code is VI.a, VI.b, VI.d, or text contains `embolism`.

3.3 Step 3: Deduplication by Drug Name

Script	Action	Output
03_deduplicate_by_drugname.py	Groups rows by <code>Drug_Name</code> to resolve conflicts from overlapping sources.	03_deduplicated_by_drugname.csv

Table 4: Step 3: Deduplication

Deduplication Logic: If the last occurrence contains "ADReCS" in the `Pattern`, the script retains the first occurrence (which typically contains the more descriptive PNEUMOTOX pattern). Otherwise, the last occurrence is retained. This prioritization is based on the fact that PNEUMOTOX is a physician-curated database, while SIDER (developed at the European Molecular Biology Laboratory - EMBL) is a comprehensive database of marketed medicines and their recorded ADRs, making both sources more reliable than ADReCS for duplicate resolution.

3.4 Step 4: Augmentation with Unlabeled Structures

Script	Action	Output
04_augment_drugcentral_smiles.py	Loads <code>drugcentral_smiles.csv</code> and constructs new rows with all 4 labels set to 0.	04_augmented_drugcentral.csv

Table 5: Step 4: Augmentation with unlabeled structures

Rationale: This adds structurally diverse molecules that lack direct evidence for the 4 target ADRs.

3.5 Step 5: Integration of Mechanistic Markers (CTDbase)

Script	Action	Output
05_integrate_ctd_markers.py	Loads ctd_mechanism_markers.csv. The CTD file uses a simplified Pattern column (e.g., plain text "ILD").	05_merged_ctd_positive_pool.csv

Table 6: Step 5: Integration of CTDbase markers

Specific Mapping:

- `Pattern == "ILD" → Label_ILD = 1`
- `Pattern == "pulmonary edema" → Label_Edema = 1`
- `Pattern == "pleural effusion" → Label_Pleural = 1`
- `Pattern == "pulmonary embolism" → Label_Embolism = 1`

Rationale: This step adds data from a reputable database (CTDbase) to enrich the positive sample pool with biologically validated mechanistic markers.

3.6 Step 6: Candidate Negative Mining and PubChem Validation

3.6.1 6.1 Isolation of Candidate Negatives

Script	Action	Output
06a_create_candidate_negatives.py	Extracts rows from the Positive Pool where all 4 labels are exactly 0.	06_candidate_negatives_raw.csv

Table 7: Step 6.1: Isolation of candidate negatives

3.6.2 6.2 PubChem API Scoring

Script	Action	Output
06_score_candidates_pubchem_api.py	Queries the PubChem PUG-View (for textual summaries) and Assay Summary APIs. Scans only adverse effects sections for curated keywords.	07_pubchem_scored_negatives.csv

Table 8: Step 6.2: PubChem API scoring

Scoring Details:

- For each compound, queries the **PubChem PUG-View** and **Assay Summary** APIs.
- Scans only the "Adverse Effects", "Toxicity", "Safety", and "Warning" sections for a curated keyword list covering the 4 target diseases.
- **Total Score** = (Keyword matches in PUG-View) + (Keyword matches in Assay Summary).

3.7 Step 7: Final Dataset Assembly

Script	Action	Output
07_assemble_final_dataset.py	Constructs the final balanced dataset by concatenating positive and negative sets.	09_balanced_pulmonary_adr_dataset.csv

Table 9: Step 7: Final dataset assembly

Assembly Details:

- **Positive Set:** All rows from `05_merged_ctd_positive_pool.csv` that have at least one `label == 1`.
- **Negative Set:** All rows from `07_pubchem_scored_negatives.csv` where the `score == 0` (i.e., no evidence of the 4 ADRs in the entire PubChem database).

4 Process Flowchart

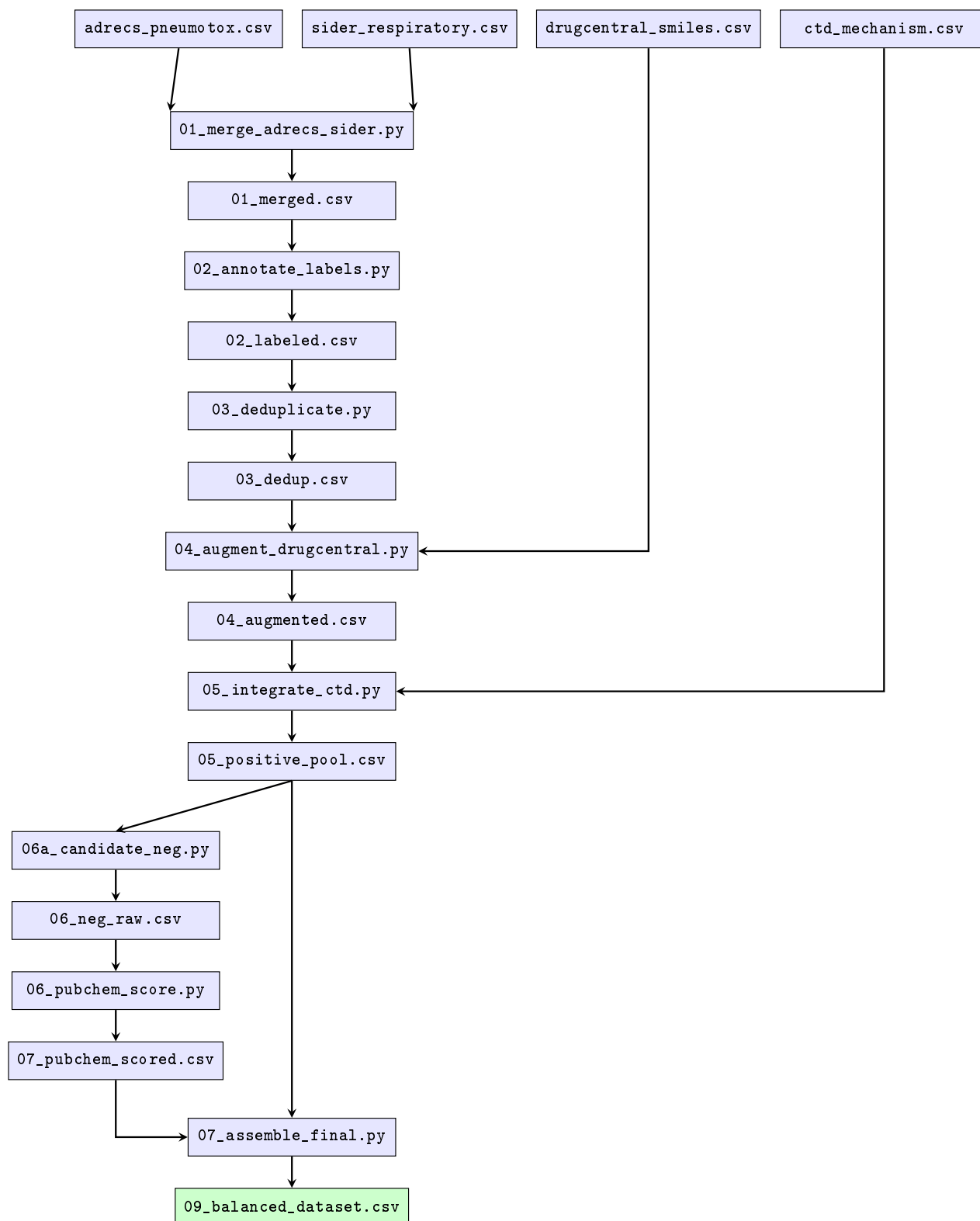


Figure 1: Complete data processing flowchart